

General Information

- Download and read the paper
- **A Survey on Wireless Indoor Localization from the Device Perspective**
- ACM Computing Surveys, vol. 49, no. 2, Article 25, June 2016.
- <https://dl.acm.org/doi/10.1145/2933232>
- (or with NTU Login)
- <https://dl.acm.org.remotexs.ntu.edu.sg/doi/10.1145/2933232>

Review of the Survey Paper

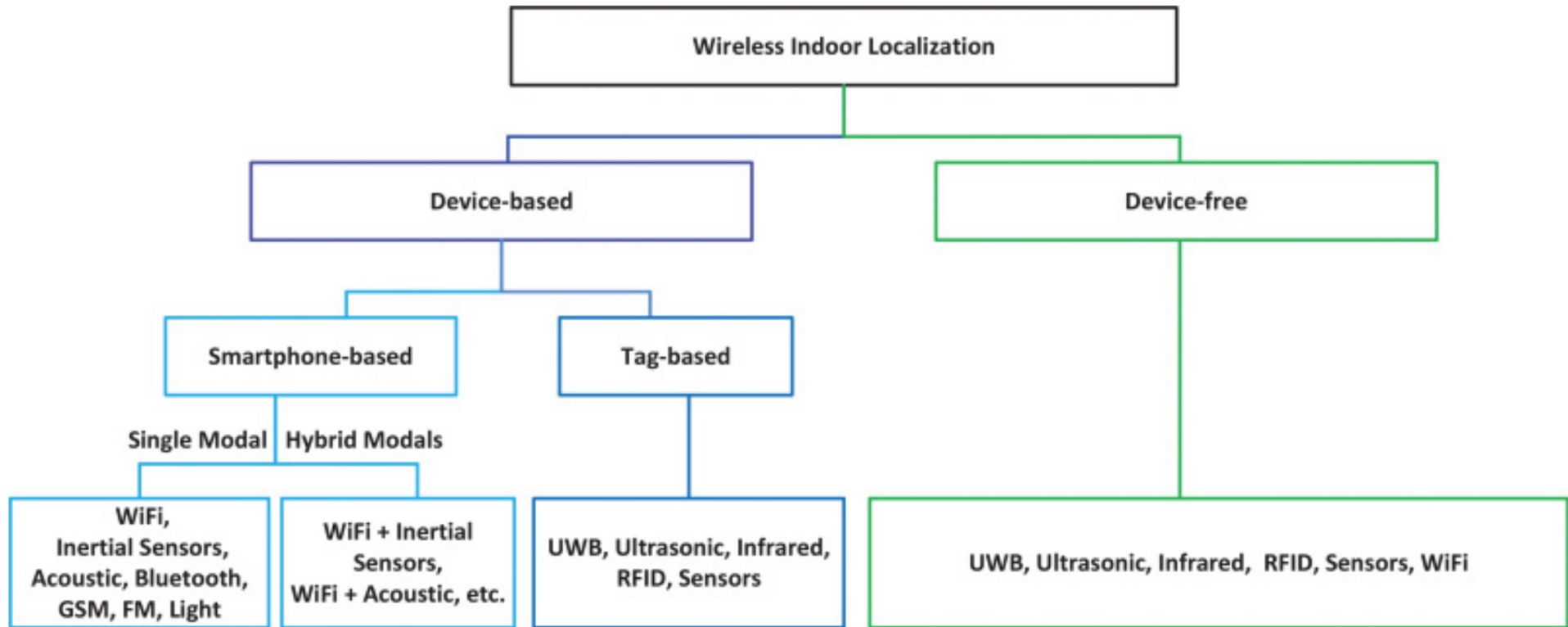
- **Indoor location-based services**



- **Global Positioning System (GPS) – Why not?**
 - Outdoors with accuracy within 3 meters
 - Unreliable in indoor due to blockage of satellite signals

Image credit: <https://bluefletch.com/the-guide-to-indoor-location-services/>

Review of the Survey Paper

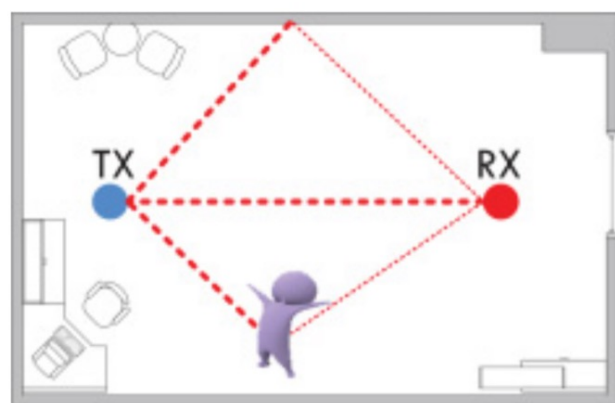


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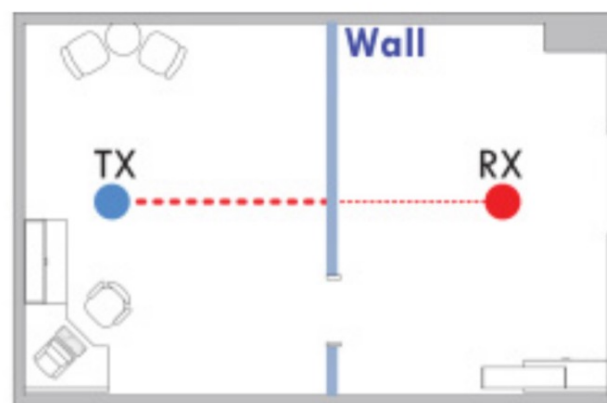
- Requirements for device-based approaches
 - Accuracy
 - Real-time
 - Cost
 - Hardware cost
 - Infrastructure cost
 - Installation cost
 - Maintenance cost
 - Energy efficiency
 - Scalability
 - Privacy

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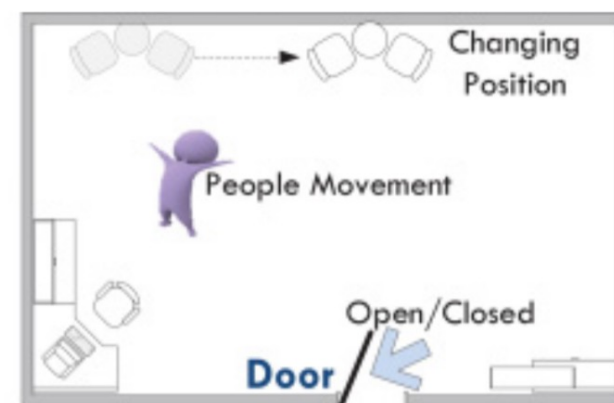
- Challenges faced by device-based approaches
 - Signal reflection
 - Signal blocking
 - Environment dynamics



(a) Signal Reflection



(b) Signal Blocking



(c) Environment Dynamic

Review of the Survey Paper

- Requirements for device-free approaches
 - Accuracy
 - Reliability
 - Real-time
 - Robustness
- Challenges faced by device-free approaches
 - Anonymity
 - Mutuality

Review of the Survey Paper

- Device-based approaches
 - Smartphone-based
 - Camera, Wi-Fi (ToA, AoA, RSS), Bluetooth, inertial sensors, acoustic, FM radio, cellular, hybrid
 - Tag-based
 - Ultra-wideband (UWB), ultrasonic, RFID, sensors

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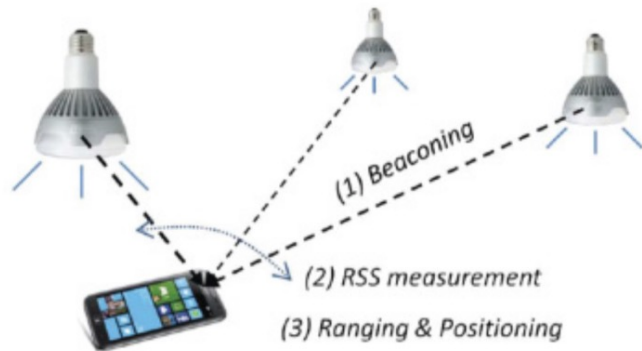


Fig. 6. Light based: Epsilon [Li et al. 2014a].

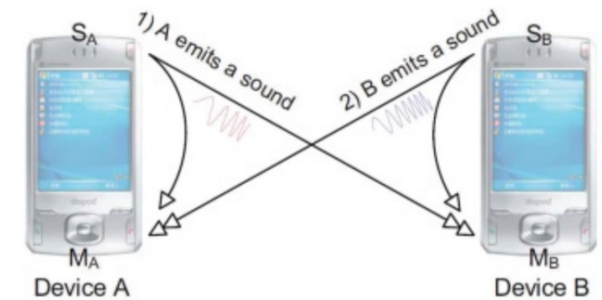
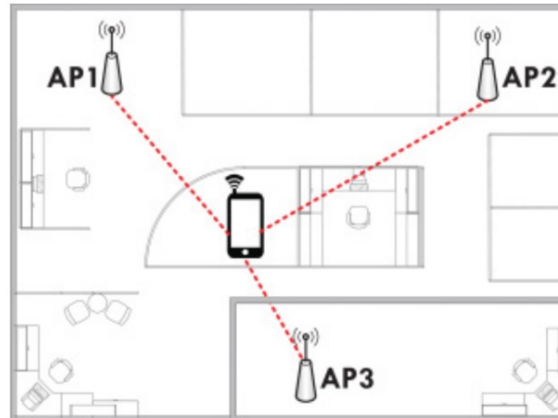


Fig. 9. Acoustic ranging: Peng et al. [2007].

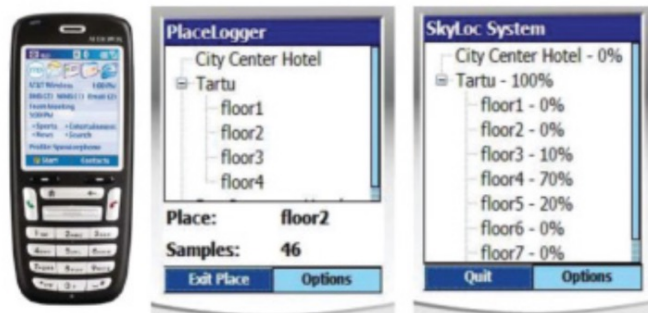


Fig. 11. GSM: SkyLoc [Varshavsky et al. 2007].

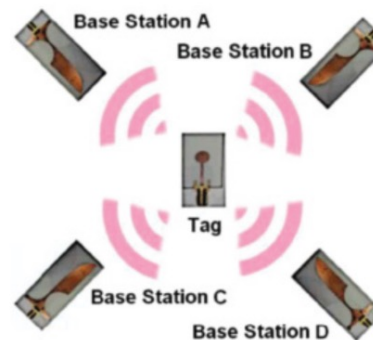


Fig. 12. UWB.

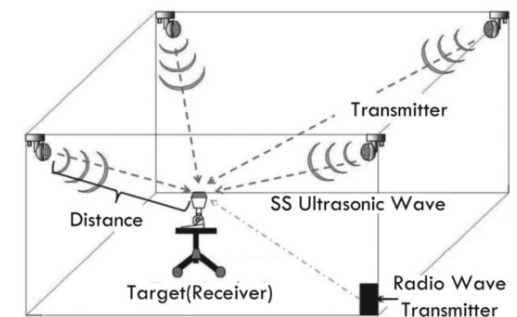


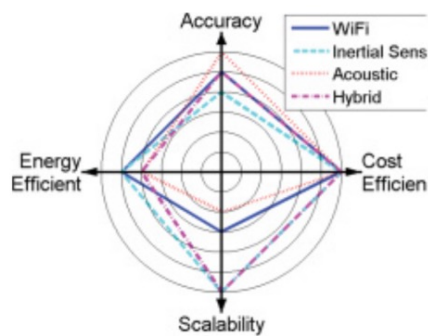
Fig. 13. Ultrasonic [Itagaki et al. 2012].

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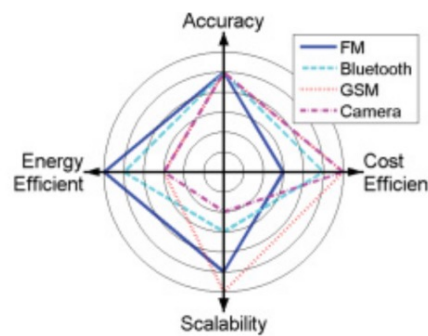
- Performance comparison of device-based approaches

Table II. Judging Criteria in Terms of Accuracy, Cost, Scalability, and Energy

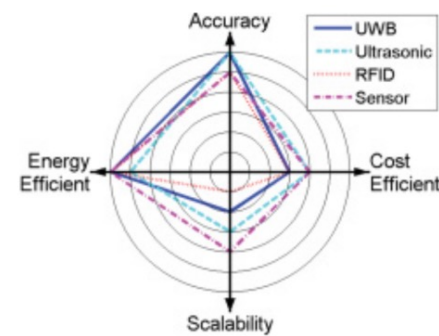
	0 (Worst)	1	2	3	4	5	6 (Best)
Accuracy	$>100m$	$51m-100m$	$16m-50m$	$7m-15m$	$2m-6m$	$0.5m-2m$	$<20cm$
Cost (US\$)	$>300k$	$101k-300k$	$31k-100k$	$10k-30k$	$6k-10k$	$1k-5k$	$<1k$
Scalability	$<10m^2$	$11m^2-50m^2$	$51m^2-100m^2$	$101m^2-500m^2$	$501m^2-1000m^2$	$1001m^2-0.1km^2$	$>0.1km^2$
Energy	$>100w$	$20w-100w$	$5w-20w$	$1w-5w$	$0.1w-1w$	$1mw-0.1w$	$<1mw$



(a) Smartphone-based
(WiFi, Inertial Sensor,
Acoustic, Hybrid)



(b) Smartphone-based
(FM, Bluetooth, GSM, Camera)



(c) Tag-based
(UWB, Ultrasonic, RFID,
Sensor)

Fig. 14. Performance comparison of all recent device-based approaches.

Review of the Survey Paper

- Device-free approaches
 - Camera, infrared, UWB radar, sonar, Wi-Fi, RFID, FM

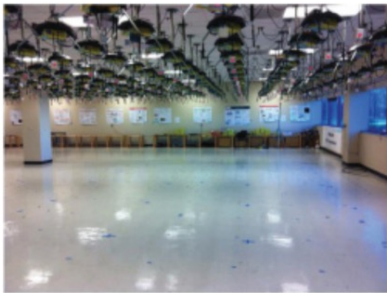


Fig. 15. Camera DfP [Xu et al. 2012].

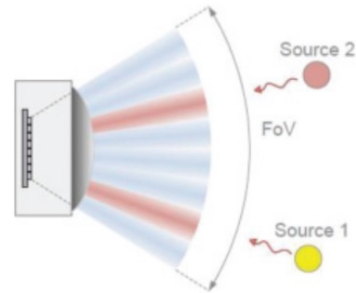


Fig. 16. Infrared DfP [Kemper and Hauschildt 2010].

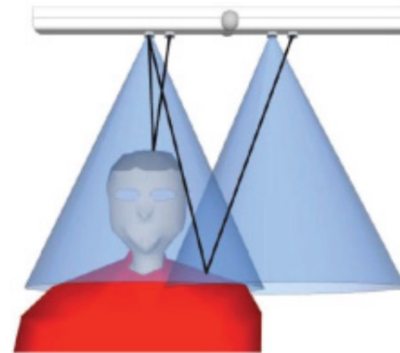


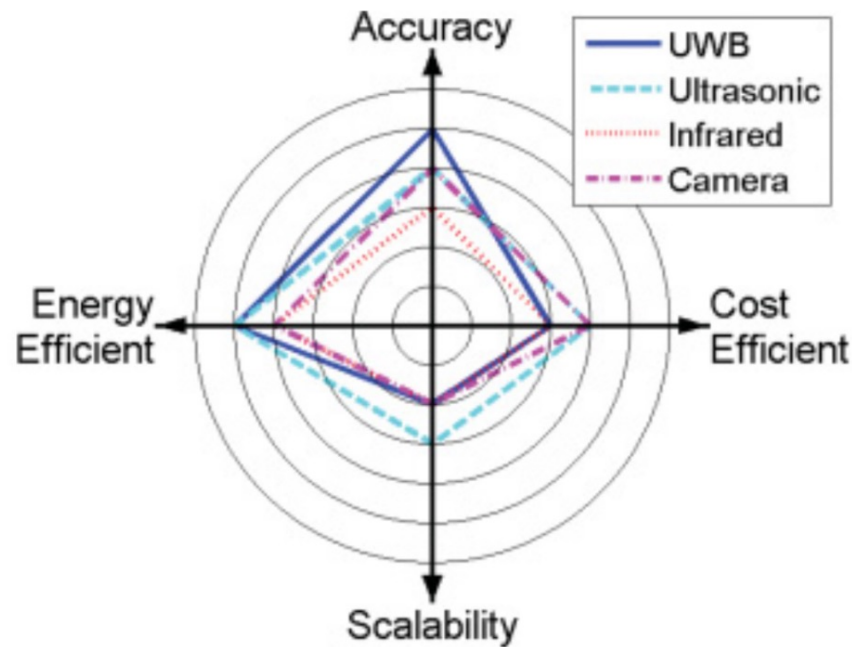
Fig. 18. Ultrasonic DfP.



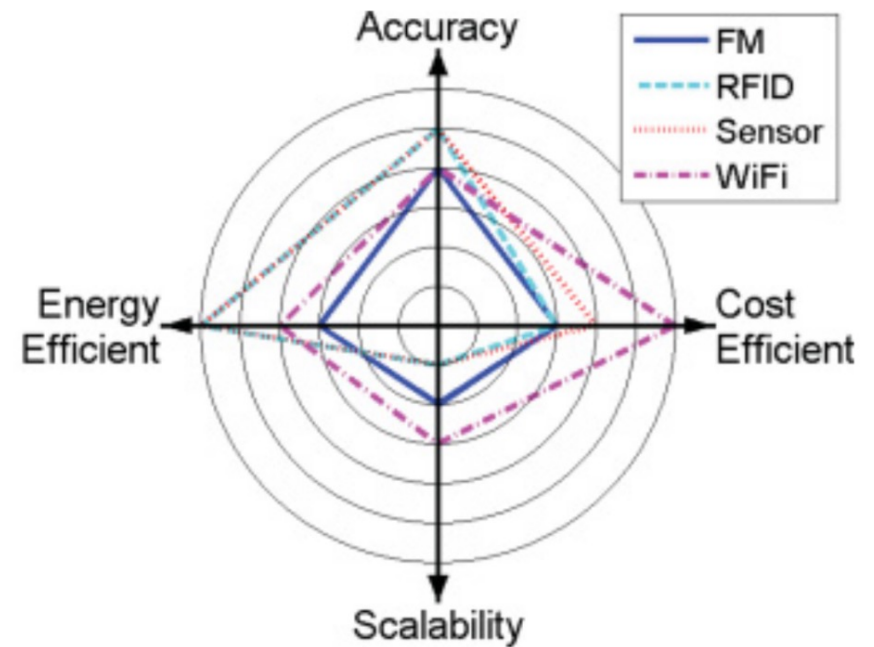
(a) Pilot overview.

Fig. 19. Wi-Fi

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(a) UWB, Ultrasonic, Infrared, SenCam



(b) FM, RFID, Sensor, WiFi

Fig. 23. Performance comparison of all recent device-free approaches.

Indoor Localization Algorithms

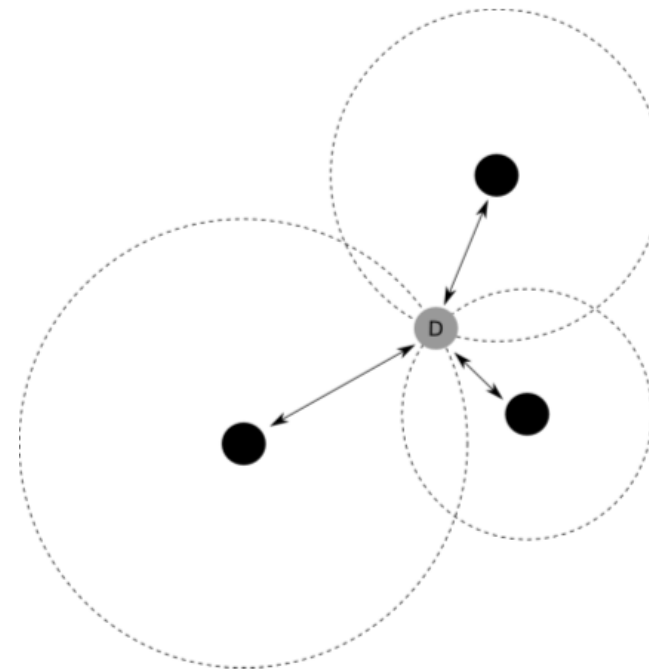
- Localization algorithms
 - Proximity
 - Lateration
 - Angulation
 - Fingerprinting

Proximity

- Simplest localization technique
- Closeness to a reference point
 - It can be used to decide whether a node is in the proximity of an anchor
 - Based on loudness, physical contact, etc
- Can be used for positioning when several overlapping anchors are available
 - Centronoid localization

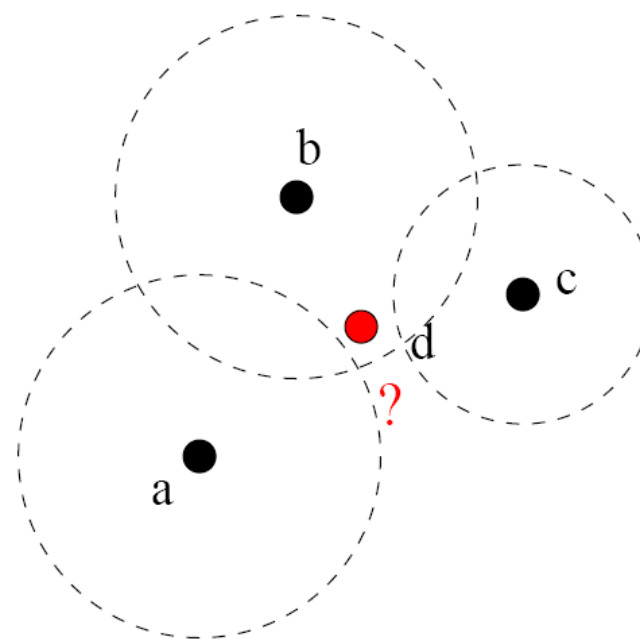
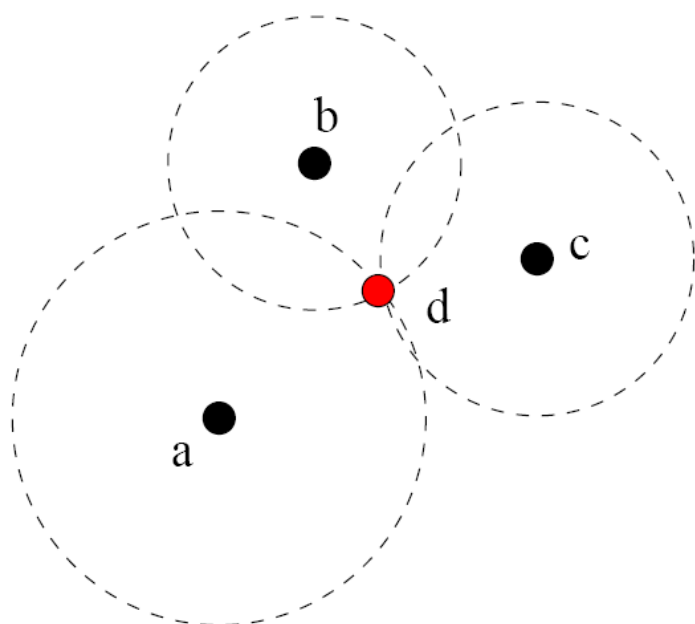
Lateralation

- Measure distance between device and reference points
- 3 reference points needed for 2D
 - Trilateration
- 4 for 3D



Noisy Distance Measurements

- Distance measurements are noisy!



Optimization Problem

- Solve an optimization problem: minimize the mean square error

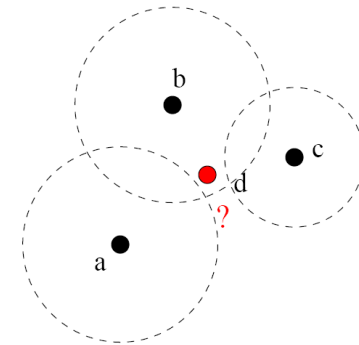
$$d^* = \arg \min_d (D(d, a) - \tilde{D}_{da})^2 + (D(d, b) - \tilde{D}_{db})^2 + (D(d, c) - \tilde{D}_{dc})^2$$

a, b, c: 2D position of anchors (known)

d: 2D position of target (unknown)

$D(x, y)$: Euclidean distance between x and y

$\tilde{D}_{xy}$: noisy distance measurement between x and y



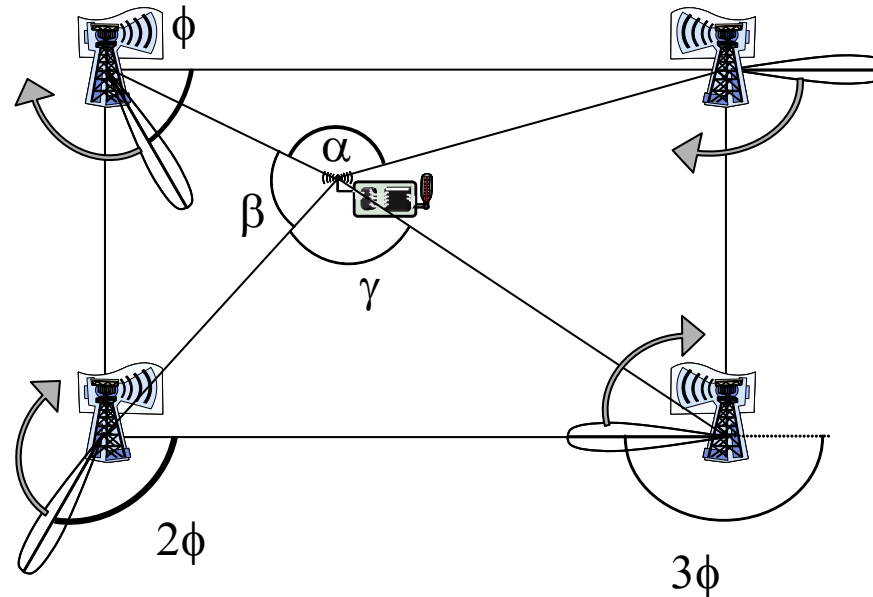
- Incorporate more anchors to reduce impact of noises

Lateralation vs Angulation

- When distances between entities are used, the approach is called lateralation
- When angles between nodes are used, one talks about angulation
- 3 reference points needed for 2D
 - Triangulation

Determining Angles

- Directional antennas
 - On the node
 - Mechanically rotating or electrically “steerable”
 - On several access points
 - Rotating at different offsets
 - Time between beacons allows to compute angles



Fingerprinting

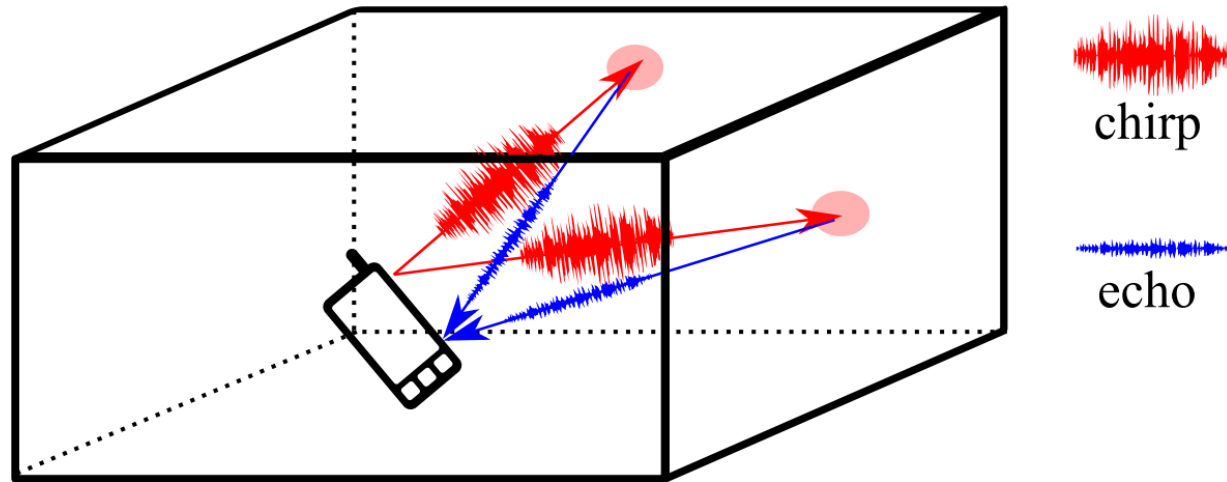
- Mapping solution
- Address problems with multipath
- Better than modeling complex signal propagation pattern

SSID (Name)	BSSID (MAC address)	Signal Strength (RSSI)
linksys	00:0F:66:2A:61:00	18
starbucks	00:0F:C8:00:15:13	15
newark wifi	00:06:25:98:7A:0C	23

Fingerprinting

- Easier than modeling
- Requires a dense site survey
- Usually better for symbolic localization
- Performance depends on
 - Spatial differentiability
 - Temporal stability

Acoustic Echo Fingerprint



- Fingerprinting
 - <https://youtu.be/5si0Cq6LzT4>
- Fingerprinting + IMU
 - <https://youtu.be/5BVTtu-Nfko>